AA-TV wiring v8.2: no wheels. 6x Deutsch DTP 4-pin (wet zone), servo leads + cables epoxy-potted into dryboxes, 14 AWG marine outside / 20 AWG on boards.

BUOY (surface, dry)

5V_PI
BT2
USB-C power bank 5 V / 3 A
GND_PI

J1

GPIO12/PWM0	32	2	5V
GPIO13/PWM1	33	1	3V3
GPIO14/UART_TXD	8	27	ID_SD_I2C0/GPIO00
GPIO15/UART_RXD	10	28	ID_SC_I2C0/GPIO01
GPIO16/SPI1_CE2	36	3	SDA_I2C1/GPIO02
GPIO17/SPI1_CE1	11	5	SCL_I2C1/GPIO03
GPIO18/SPI1_CE0/PCM_CLK/PWM0	12	7	
GPIO19/SPI1_MISO/PCM_FS	35	29	GPCLK0/GPIO04
GPIO20/SPI1_MOSI/PCM_DIN/PWM1	38	31	GPCLK1/GPIO05
GPIO21/SPI1_SCLK/PCM_DOUT	40	33	GPCLK2/GPIO06
GPIO22/SDIO_CLK	15	26	CE1_SPI0/GPIO07
GPIO23/SDIO_CMD	16	24	CE0_SPI0/GPIO08
GPIO24/SDIO_DAT0	18	21	MISO_SPI0/GPIO09
GPIO25/SDIO_DAT1	22	19	MOSI_SPI0/GPIO10
GPIO26/SDIO_DAT2	37	23	SCLK_SPI0/GPIO11
GPIO27/SDIO_DAT3	13		
GND	6		

Raspberry Pi 4 (buoy)
GND_PI

Pi 4 talks to the Pico over USB serial (/dev/ttyACM0) and powers it from USB 5 V. Reset the Pico by power-cycling the Pi USB ports (uhubctl).

TETHER

W1
USB-A
micro-B
+ USB data
5V_PI
GND_PI
VBUS_PICO
GND

USB-A to micro-B, ~3 m

Vinyl air tube from the buoy; USB cable runs inside it. USB plugs are dry (buoy, Pico).

PEN1
A1: air-tube fitting (USB inside)

MAIN DRYBOX (upper, on hull)

BOARD A: Pico solder-board

R1 220 GP12 M4_PWM
R2 220 GP13 M5_PWM
R3 220 GP14 M6_PWM
R4 220 GP15 M7_PWM
R5 220 GP16 THR_PWM
R6 220 GP19 PUMP_PWM

Raspberry Pi Pico A1

VBUS_PICO	39		
VSYS	40		
VBUS			
+3V3		36	3V3
RUN	30		
3V3_EN	37		
GPIO0	1		
GPIO1	2		
GPIO2	4		
GPIO3	5		
GPIO4	6		
GPIO5	7		
GPIO6	9		
GPIO7	10		
GPIO8	11		
GPIO9	12		
GPIO10	14		
GPIO11	15		
GPIO12	16		
GPIO13	17		
GPIO14	19		
GPIO15	20		
GPIO16		21	GP16
GPIO17		22	
GPIO18		24	
GPIO19		25	GP19
GPIO20		26	
GPIO21		27	
GPIO22		29	
ADC_VREF		35	
GPIO26_ADC0		31	LEAK_MAIN
GPIO27_ADC1		32	LEAK_REAR
GPIO28_ADC2		34	
AGND		33	GND
GND		3	GND

220 ohm in series on every PWM line that leaves the drybox.

TA2
+3V3
GND
LEAK_MAIN
Main leak sensor

TA1

M4_PWM	1
M5_PWM	2
M6_PWM	3
M7_PWM	4
THR_PWM	5
H1_SPARE6	6
H1_SPARE7	7
PUMP_PWM	8
LEAK_REAR	9
+3V3	10
GND	11
GND	12

H1 out (14 AWG)

All Board A terminals: 5.08 mm pitch (0.2 in, fits the breadboard grid). TA1 pins 6-7: spare H1 conductors.

PEN2
A2: H1 cable, epoxy-potted

REAR DRYBOX (inside open rear endcap)

BOARD B: servo / signal board

TB1

M4_PWM	1
M5_PWM	2
M6_PWM	3
M7_PWM	4
THR_PWM	5
H1_SPARE6	6
H1_SPARE7	7
PUMP_PWM	8
LEAK_REAR	9
+3V3	10
GND	11
GND	12

H1 in (14 AWG)

TB7

SERVO_7V5	1
SERVO_7V5	2
GND	3
GND	4

7.5 V in (2x14 AWG per pole)

SERVO_7V5
C1
1000uF 16V
GND

TB2

M4_PWM	1
SERVO_7V5	2
GND	3
M5_PWM	4
SERVO_7V5	5
GND	6

Rudder servo leads (potted)

Board B:
- 7.5 V + GND rails: strip of 14 AWG soldered along strips
- Servo + / - jumpers: 14 AWG
- Signal jumpers: 20 AWG
- No 12 V on the breadboard
- TB1 pins 6-7: spare, unused

TB3

M6_PWM	1
SERVO_7V5	2
GND	3
M7_PWM	4
SERVO_7V5	5
GND	6

Elevator servo leads (potted)

TB4

THR_PWM	1
GND	2

Thruster ESC signal (to D3)

TB5

PUMP_PWM	1
GND	2

Pump ESC signal (U3)

TB6

+3V3	1
GND	2
LEAK_REAR	3

Rear leak sensor

PEN3
R2: signal cables potted (H1, 4 servo leads, THR sig)

Off-board, in rear drybox

BB1
VBAT
Bus bars (OONO 2x6 screw-clamp module, one row per pole, 30 A); in: D1 trunk, two 14 AWG per pole out: buck U1, pump ESC U3, D3 thruster ESC feed (1 spare per row)
+12 V row (OONO 2x6, 30 A)

BB2
GND
GND row (OONO 2x6, 30 A)

U1
BUCK
VBAT 1 VIN+ VOUT+ 3 SERVO_7V5
GND 2 VIN- VOUT- 4 GND
12 V -> 7.5 V buck (>=5 A)

U3
VBAT 1 BAT+ M+ 6 PUMP_M+
GND 2 BAT- M- 7 PUMP_M-
PUMP_PWM 4 SIG
GND 5 SIG_GND ESC
Pump ESC (brushed, reversible)

Pump ESC sits dry, so any brushed ESC with reverse works. BEC wire cut. Motor leads -> R3 -> D2 pins 1-2 (extend with 14 AWG if short).

PEN4
R3: power cables potted (trunk, thruster feed, pump)

WET ZONE (flooded hull, open rear endcap, splash areas): Deutsch DTP 4-pin connectors live here

30 A ATC, waterproof holder (>=30 A)
F1
VBAT
BT1
3S LiPo (wet, no box)
SW1
GND
Kill switch, low side, >=30 A DC

Battery side: battery + -> F1 -> split -> D1 pins 1-2; battery - -> SW1 -> split -> D1 pins 3-4. Two contacts per pole (2 x 25 A). Each split: one sealed Y-splice (adhesive heat shrink) or two ring terminals on a stud. Charge through a DTP04-4P -> XT60 adapter.

D1A D1B
D1 battery DTP06-4S DTP04-4P
D1: 1-2 +12 V, 3-4 GND (2 x 25 A per pole).

D4A D4B
D4 H1 1-4 DTP06-4S DTP04-4P

D5A D5B
D5 H1 5-8 DTP06-4S DTP04-4P

D6A D6B
D6 H1 9-12 DTP06-4S DTP04-4P
H1 main <-> rear: 12 x 14 AWG over D4, D5, D6. Pins 8-7 spare.

M4 Rudder servo
M5 Rudder servo
M6 Elevator servo
M7 Elevator servo

Servos: 2x Hitec DB777WP (4.5 A stall @7.4 V), 2x Savox SW-1250MG (2.4 A). Servo leads: potted straight into the rear drybox at R2, no connector, BACKUP if a lead can't reach the box: one MGYOO 20 AWG 3-pin waterproof connector per servo, inline in the wet zone.

D3A D3B
D3 thruster DTP06-4S DTP04-4P
D3: 1 +12 V, 2 GND, 3 PWM (yellow), 4 signal GND (white). All 14 AWG.

U2 BLDC ESC
MT1 BLDC
TD1.2 thruster
TD1.2 ESC (waterproof, wet)
Factory thruster cable stays on the ESC.

D2A D2B
D2 pump DTP06-4S DTP04-4P
D2: 1 pump +, 2 pump -; 3-4 dummy loop.

M3
Ballast pump, 12 V brushed reversible (TBD, ~10 A)

Deutsch DTP: size 12 contacts, 12-14 AWG, 25 A each. Every cavity is filled: D2 pins 3-4 get a dummy wire loop. Source side (battery / drybox) gets the DTP06 plug with socket contacts.

Wire: 14 AWG marine for everything outside the boxes, every DTP contact and all power; 20 AWG hookup inside the boxes.
Only connectors: 6x Deutsch DTP 4-pin in the wet zone + 5.08 mm screw terminals / bus bars inside the boxes.
Two solder-boards (Board A main drybox, Board B rear drybox). Cables epoxy-potted through drybox walls.

PIE AA-TV
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Title: AA-TV electrical v8.2: no wheels, 6x Deutsch DTP 4-pin, servos potted

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